

1. Sol

* Electric field lines show the strength and direction of electric force around a charge. Around a positive charge, lines move outward, while around a negative charge, lines move inward.

* When two opposite charges are near each other, field lines connect from positive to negative, showing attractive force. For like charges, field lines spread apart, showing repulsive force.

* Therefore, electric field line patterns help us visually understand the direction, strength, attraction and repulsion of electric forces between charges.

2. * Electric field representation is better for systems with multiple charges because it gives the effect of all charges at a point directly.

* It is simple and more efficient than calculating separate force vectors for every charge interaction.

3. * A capacitor charges and discharges to create time delay in electronic circuit.

* In a RC circuit, the timing depends on the values of resistance and capacitance.

* Stable dielectric materials keep capacitance constant, giving accurate and consistent timing.

4. * Proper capacitor selection improves device reliability by ensuring safe operation under different voltages and preventing break down or failure

* Engineers consider dielectric strength, durability, performance and cost to achieve safe, efficient and long lasting electronic devices.

5. * Identify the positions and magnitudes of all fixed charges in the system.

* Calculate the electric field produced by each charge at the required point using Coulomb's law.

* Add all individual electric fields vectorially to obtain the net field intensity accurately.

6. * Discrete summation is suitable for a small number of charges, but calculations become complex and time consuming as the number of charges increases.

* For a large system, continuous charge distribution is more efficient because integration gives the overall electric field in a simple and systematic way.

7. * Identify the required equivalent capacitance and the values of available capacitors

* Use parallel combination to increase capacitance ($C = C_1 + C_2 + \dots$) and series combination to decrease capacitance ($\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$)

* Combine capacitors step by step and verify the final equivalent capacitance to ensure the correct arrangement is obtained.

8. * Increasing capacitance improves energy storage and voltage stability, which increases system efficiency and performance.

* However, larger capacitance increases size, cost, and heat generation, reducing scalability and making system design more difficult.

9. * According to Gauss's law, the total electric flux through a closed surface is equal to the enclosed charge divided by permittivity of free space



$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

For a charged spherical shell,

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

$$A = 4\pi r^2$$

electric flux is $\Phi = E \times A$

$$\Phi = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \times 4\pi r^2$$

$$\boxed{\Phi = \frac{Q}{\epsilon_0}}$$

* Thus, the radius r cancels out, showing that the electric flux remains constant for any gaussian surface enclosing the same charge.

10. * In electrostatic equilibrium, the electric field inside a grounded conducting chamber should be zero.

* Small gaps, joints or cracks in the conducting walls allow external fields to enter the chamber.

* Poor conductivity of the material can prevent proper redistribution of charges

* Improper grounding may leave induced charges on the chamber surface, causing internal electric fields.

* Hence, imperfections in the conductor and faulty grounding disturb electrostatic shielding and produce non-zero electric field inside the chamber

11. Electric flux through a flat surface in a uniform electric field is given by:

$$\Phi = \vec{E} \cdot \vec{A} = EA \cos \theta$$

E = magnitude of electric field

A = area of the surface

θ = angle between electric field and the area vector

* The fluctuating values occur because the angle is measured incorrectly with the surface instead of normal. This leads to wrong projection of area and incorrect use of $E \sin \theta$ instead of $E \cos \theta$

* Flux is maximum when the field is perpendicular to the surface and zero when parallel

* Hence, the error is due to misunderstanding of surface orientation and flux direction

12. * The torque on an electric dipole in a uniform electric field

$$\tau = pE \sin \theta$$

Where

$p = qd$ is the dipole moment

E = Electric field strength

θ = angle between dipole axis and electric field

* The torque is depend on angular position:

i) Torque is maximum at $\theta = 90^\circ$

ii) Torque is ~~in~~ zero at $\theta = 0^\circ$ or 180°

* Thus, the dipole tends to rotate and align itself parallel to the electric field. Greater angular displacement produces greater rotational tendency up to 90°

13. * To minimize energy consumption while maintaining moderate and uniform spacing to create a nearly uniform electric field.

* The relation between electric field & potential is

$$E = -\frac{dV}{dx}$$

* work done is

$$W = q \Delta V$$

* If charges are placed too close, the electric field becomes very strong, increasing work and energy usage.

* If placed too far apart, the potential differential becomes too small for sensing

* Hence, an optimal configuration uses evenly spaced charges to produce a stable electric field with sufficient potential difference and reduced energy expenditure.

14.

* Resistance of a wire is $R = \rho \frac{L}{A}$

When the wire is doubled, $R' = 2R$

* If temperature increases, resistivity ρ also increases for copper, so resistance becomes greater than $2R$

* Using ohm's law, $V = IR \Rightarrow I = \frac{V}{R}$

current inversely decreases with resistance

* Drift velocity is $v_d = \frac{I}{nAe}$

since, current decreases, drift velocity also decreases

* Thus, doubling the length increases resistance, temperature rise further increases resistivity and both effects reduce current and drift velocity under constant voltage condition

15. * Electric field and potential are related by

$$E = -\frac{dv}{dx}$$

* Uneven spacing of equipotential surfaces means the electric field is non-uniform. The inconsistency is caused by incorrect modeling of discrete and continuous charge distributions and neglect of superposition effects.

* This produces wrong electric field values, causing charged particles to deviate from expected paths and give incorrect estimation of stored electrical energy

16. * Resistance of a transmission wire is given by

$$R = \rho \frac{L}{A}$$

* Low resistivity materials such as copper or aluminium reduce resistance and power losses.

* Increasing wire thickness increases area A which further decrease resistance and improves current carrying capacity.

* Power loss is $P = I^2 R$. Also, lower resistance gives better efficiency.

Trade-offs

- i) Copper - better conductivity and strong but expensive and heavy
- ii) Aluminium - cheaper and lighter, but conduct less
- iii) Thicker wire $\rightarrow$ lower power loss, but higher cost and weight
- iv) Higher temperature $\rightarrow$ increases resistance and reduces efficiency

* Hence, a lower resistance conductor with moderate thickness gives the best balance of efficiency, cost and reliability for power transmission.

17. In the analogy:

* Each person represents a magnetic domain

* Random facing directions represent an unmagnetized material.

* Music represents the external magnetic field

* People aligning in one direction represents magnetization

When the music stops, thermal motion and internal disturbances cause domains to return to random orientations, so the material loses its magnetization.

18. * The simpler assumption is that magnetism occurs only due to visible current in conductors.

* Magnetism also comes from electron spin and orbital motion inside atoms. Alignment of these atomic magnetic moments produces magnetism even without visible current flow.

19. To obtain the required equivalent capacitance:

* Use parallel combination to increase capacitance

$$C_p = C_1 + C_2 + C_3 + \dots$$

* Use series combination to decrease capacitance

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

* The engineer should first compare the required capacitance with available values, then choose suitable series or parallel arrangements while considering space, voltage rating and circuit performance constraints.

20. * Use a capacitor with sufficient capacitance and suitable voltage rating to store and supply charge during sudden load changes.

* Choose compact, low-cost capacitors that provide fast response and good stability while fitting the design space constraint.

21. * Electric charges can exist independently, so isolated charges are observed in electrostatics.
- * When a magnet is cut, each piece still contains both a north and a south pole.
- * Since, no isolated magnetic pole has ever been found and magnetic field lines form closed loops, magnetic poles always occur in pairs.
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22. * The first claim is incorrect because magnetic fields are produced by moving charges or magnetic materials.
- * Hence, magnetic field cannot exist independently without a physical source.
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23. * Capacitors charge and discharge by storing and releasing electrical energy.
- * proper capacitor placement filters noise and stabilizes voltage, improving signal clarity.
- * If placed incorrectly, charging and discharging introduce delays, affecting high frequency signal transmission & reducing performance.
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24. * capacitors are suitable because they can charge and discharge quickly, providing instant bursts of energy.
- * They improve responsiveness but not suitable for long term energy storage.

25. The relationship between magnetic flux density (B) and magnetic field intensity (H) is,

$$B = \mu H$$

Given: output (B) doubles for the same input (H) when medium is changed

$$\therefore B_2 = 2B_1$$

$$\mu_2 H = 2(\mu_1 H)$$

$$\boxed{\mu_2 = 2\mu_1}$$

Given: $H = 50 \text{ A/m}$

$$B_2 = \mu_2 H = (2\mu_1)(50)$$

$$\boxed{B_2 = 100 \mu_1}$$

Result:

* Governing relationship: $B = \mu H$

* Hidden parameter: Permeability (μ)

* Output in second medium at $H = 50 \text{ A/m}$ is $B_2 = 100 \mu_1$.

26.

$$U = \frac{1}{2} L I^2$$

where, $U \rightarrow$ stored energy

$L \rightarrow$ Inductance

$I \rightarrow$ current

* Stored energy is

i) directly proportional to L

ii) proportional to square of I

* So, current (I) has greater influence on energy. For eg, doubling I increases energy 4 times, while doubling L only doubles it

* conclusion:

current has the stronger effect, but due to thermal constraints, the safer modification is to increase inductance (L).

27. * Magnetic flux is proportional to permeability and field intensity.
 $B = \mu H$

* High permeability (μ) gives better magnetic shielding.

* Material A provides stronger shielding but is heavy and costly.

* Material B is lightweight and portable but offers lower shielding performance.

* Best approach: use a thin layer or combination of material A and B to achieve 80% field reduction while balancing shielding, weight and cost.

28. * Induced emf is given by Faraday's Law:

$$\epsilon = -N \frac{d\Phi}{dt}$$

* The induced emf depends on the rate of change of magnetic flux (Φ).

* The negative sign represents Lenz's law which states that the induced emf opposes the change causing it.

* When the primary current changes, the magnetic flux changes inducing emf in the secondary coil.

* If the current is constant, $\frac{d\Phi}{dt} = 0$

$$\therefore \epsilon = 0$$

* Hence, no induced emf is produced with constant current because the magnetic flux remains unchanged.

29. * Electromagnetic force is given by: $F = BIL$
- * Force increases directly with both magnetic field strength (B) and rod length (L)
 - * Structural reinforcement increases the rod's ability to withstand higher forces without elastic deformation, allowing safer operation.
 - * Increasing B or L helps to achieve the required 6N force, but a longer rod may reduce mechanical stability.
 - * $\therefore$, The best design is the one with strong structural reinforcement and a higher magnetic field because it can produce the required force of 6N or more while preventing the rod from bending or deforming
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30. * Ohm's law ($V = IR$) is valid only when temperature and physical conditions remains constant.

- * As voltage increases, the semiconductor heats up due to internal power dissipation

- * The rise in temperature changes the material's resistance.

- * Semiconductors are non-ohmic devices, so, current rises rapidly at higher voltages.

- * Hence the deviation is due to temperature dependent resistance and semiconductor properties.

31. * Magnetic field around a long straight wire is

$$B = \frac{\mu_0 I}{2\pi r}$$

- * Magnetic field increases with current (I) and decreases with distance (r)

- * If current is doubled, the magnetic field doubles, if distance is doubled, the magnetic field becomes half.
- * With two identical wires, the resultant field is found using the principle of superposition (vector addition of individual fields)
- * Therefore, Distance has a greater impact on field strength.

32. * Current density is related to drift velocity by

$$J = nqvd$$

where

$J \rightarrow$ current density

$n \rightarrow$ charge carrier density

$q \rightarrow$ electron charge

$vd \rightarrow$ drift velocity

* $\therefore$ Current $= I = JA = nqAvd$ [From $J = \frac{I}{A}$]

* Increasing the electric field increases vd , which increases J and hence the current also increases.

* Thus, electron drift cause the observable current in the conductor.

33. * A time-varying electric field between capacitor plates produces a magnetic field even without conduction current

* Maxwell introduced displacement current to explain this magnetic effect.

* Displacement current acts like real current in generating magnetic fields, ensuring continuity of current in the circuit,

34. * In free space, a changing electric field produces a magnetic field and a changing magnetic field produces an electric field.
* ~~These~~ These mutually perpendicular fields continuously regenerate each other, allowing electromagnetic waves to ~~propagate~~ propagate through free space without any medium.

35. * The object appears displaced because light refracts when entering and leaving the glass slab.
* Due to slab's refractive index, light changes speed and bends inside the glass.
* The apparent displacement is governed by the principle of refraction (Snell's law).

36. * The defect is myopia (short-sightedness).
* It is corrected using a concave lens, which helps focus parallel rays on the retina.

37. * A changing magnetic field induces an emf in the coil.
* The induced emf produces a current in the circuit.
* According to Lenz's law, the induced current flows in a direction that opposes the change in magnetic flux causing it.

38. * Maxwell's equations combine the laws of electricity and magnetism into a single unified theory of electromagnetism.
* This theory explains field interactions and predicts the existence and propagation of electromagnetic waves.

39. * Light is confined in the core by Total Internal Reflection (TIR).

* This occurs because the core has a higher refractive index than the cladding.

* For efficient transmission, the angle of incidence must be greater than the critical angle.

40. * The ray bends toward the normal because water has a higher refractive index than air.

* This behaviour is explained by Snell's law of refraction.

41. * Stored energy in a capacitor is given by: $U = \frac{1}{2} CV^2$

Higher capacitance and voltage increases energy storage.

* Energy density in the Electric field is: $u = \frac{1}{2} \epsilon E^2$

Energy density increases with the square of electric field strength

* A high-permittivity dielectric (ϵ) increases capacitance and allows more energy to be stored.

* Very high Electric fields can cause dielectric breakdown and insulation failure

* Use high-permittivity, high-breakdown strength dielectric for safe and efficient storage.

42. Young's Double Slit Experiment

* Fringe spacing is given by: $\beta = \frac{\lambda D}{d}$ where $\lambda \rightarrow$ wavelength
 $D \rightarrow$ screen distance
 $d \rightarrow$ slit separation

* Fringe spacing increases with wavelength and screen distance

* Fringe spacing decreases with slit separation.

- * Clear fringes require coherent monochromatic light and proper slit alignment.
- * Vibrations and misalignment reduce fringe visibility. a stable setup is needed for distinct interference patterns.

43. * Oscillating currents generate changing Electric (E) and magnetic (B) fields, forming electromagnetic waves.

- * Energy flow is given by Poynting vector: $\vec{S} = \vec{E} \times \vec{H}$
- * The direction of energy propagation is perpendicular to both Electric and magnetic fields.
- * The magnitude of energy transfer depends on the strengths of the electric and magnetic fields.
- * As distance increases, wave intensity decreases, reducing power-transfer efficiency.

44. * Light reflects from the top and bottom surfaces of the thin film, producing interference.

- * The interference depends on the film thickness (t), refractive index (n) and wavelength (λ)
- * Destructive interference reduces reflection, while constructive interference increases reflection.
- * Different wavelength produces different colors.
- * To minimize reflection for a wavelength λ , use $t = \frac{\lambda}{4n}$
(quarter-wave coating)

45. * Maxwell's curl equations are combined to derive the wave equation.

- * In free space, $\nabla^2 E = \frac{1}{\mu_0 \epsilon_0} \frac{\partial^2 E}{\partial t^2}$
- * A similar wave equation is obtained for the magnetic field B
- * The wave speed is: $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$
- * Since the calculated wave speed c is equal to the speed of light, it proves that light is an electromagnetic wave.

46. * A population inversion must be achieved with more atoms in the excited state than the ground state.

* An external pumping source is used to excite ~~all~~ atoms.

* Excited atoms release photons through stimulated emission.

* A metastable state helps maintain population inversion for a longer time.

* Mirrors inside the laser reflect the light repeatedly, amplifying it and producing a continuous, coherent laser beam.

47. * A changing electric field produces a magnetic field.

* A changing magnetic field produces an electric field.

* These fields continuously generate each other, creating a self-sustaining electromagnetic wave.

* The electric and magnetic fields are perpendicular to each other.

* The wave travels in ^a direction perpendicular to both fields, carrying energy through space without a ~~nuclear~~ medium.

48. * Linear Polarization - Occurs when the phase difference is 0° or 180° . Light vibrates in a straight line.

* Circular Polarization - Occurs when the phase difference is 90° & amplitudes are equal. Light rotates in a circle.

* Elliptical Polarization - Occurs for other phase differences or unequal amplitudes. Light traces an ellipse.

* Phase Difference effect - Changing the phase difference changes the polarization states from linear to circular or elliptical.

* Practical use - Different polarization patterns help to identify stress, strain and defects in transparent materials.
